Week 11 APP DEV-2

2024 Dec22 ET

1. [Source Q15; MCQ]

Which of the following best describes the difference between @cache.memoize() and @cache.cached() in Flask-Caching?

(a)

@cache.memoize() caches function results based on arguments, while @cache.cached() caches entire views.

(b)

@cache.cached() caches function results based on arguments, while @cache.memoize() caches entire views.

(c)

Both cache entire views, but @cache.memoize() uses Redis by default.

(d)

Both methods are identical but use different syntax.

[Source ans: a]

2025 Apr13 ET

2. [Source Q5; MCQ]

Which of the following is NOT a common performance metric measured by Google Lighthouse?

(a) First Contentful Paint (FCP)

(b) Time to Interactive (TTI)

(c) Total Blocking Time (TBT)

(d) Database Query Time (DQT)

[Source ans: d]

3. [Source Q15; MCQ]

Consider the following Flask application (server.py):

server.py

```python
from flask import Flask
from flask_caching import Cache
import time

config = {
    "DEBUG": True,
    "CACHE_TYPE": "RedisCache",
    "CACHE_REDIS_URL": "redis://localhost:6379/1",
    "CACHE_DEFAULT_TIMEOUT": 300
}

app = Flask(__name__)
app.config.from_mapping(config)
cache = Cache(app)

@app.route('/data/<item>')
@cache.cached(timeout=100)
def get_data(item):
    time.sleep(25)
    return f"Data for {item}"

if __name__ == '__main__':
    app.run()
```

The application runs on http://localhost:5000.
A client makes two consecutive requests to http://localhost:5000/data/apple within 50 seconds
What will be the approximate absolute difference between their latencies?

(a) 25 seconds

(b) 35 seconds

(c) 0 seconds

(d) 15 seconds

[Source ans: a]

4. [Source Q22; MCQ]

Consider the following Flask application (server.py) and an HTML file named index.html:

server.py

```python
from flask import Flask
import time

app = Flask(__name__)

@app.route("/api/slow1")
def slow_api_1():
    time.sleep(15)
    return "Response from Slow API 1", 200, {'Access-Control-Allow-Origin':
'*'}

@app.route("/api/slow2")
def slow_api_2():
    time.sleep(25)
    return "Response from Slow API 2", 200, {'Access-Control-Allow-Origin':
'*'}

if __name__ == "__main__":
    app.run(threaded=False)
```

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Flask Fetch Example</title>
</head>
<body>
    <h1>Fetching API Data...</h1>

    <script>
        fetch("http://127.0.0.1:5000/api/slow1")
```

```
            .then(res => res.json())
            .then(data => console.log("API 1 Response:", data));

        fetch("http://127.0.0.1:5000/api/slow2")
            .then(res => res.json())
            .then(data => console.log("API 2 Response:", data));
    </script>
</body>
</html>
```

If you open index.html in a browser, approximately how long will it take for the second fetch call (http://127.0.0.1:5000/api/slow2) to complete and log the data to the console?

(a) 15 seconds

(b) 25 seconds

(c) 10 seconds

(d) 40 seconds

[Source ans: d]

2025 Aug31 ET

5. [Source Q4; MCQ]

In Flask, what is the main benefit of using the @cache.cached() decorator?

(a) It stores output in session storage

(b) It enables token authentication

(c) It sends compressed response headers

(d) It skips route execution for repeated inputs

[Source ans: d]

6. [Source Q7; MSQ]

Which of the following factors directly affect the speed and performance of a web page load?

(a) The number of CSS selectors in the stylesheet

(b) Use of semantic HTML tags

(c) Number of HTTP requests made

(d) File size of resources (like images, JS)

[Source ans: c, d]

7. [Source Q11; MCQ]

What is the main advantage of using caching in a Flask-based web application?

(a) It eliminates the need for client-side rendering

(b) It reduces server load by reusing responses for repeated requests

(c) It encrypts data transmission between server and client

(d) It avoids the need for database interactions completely

[Source ans: b]

8. [Source Q17; MSQ]

Which factors directly impact the Lighthouse "Performance" score?

(a) First Contentful Paint (FCP)

(b) Time to Interactive (TTI)

(c) Layout Shift

(d) HTTPS usage

[Source ans: a, b, c]

2025 Dec18 ET

Shared context (source Q26)

Consider the Vue application.

```
<div id="app">
   <p>{{ expensive }}</p>
   <button @click="calculate">Calculate</button>
 </div>
 <script>
   new Vue({
     el: "#app",
     data: {
       a: 1,
       b: 2,
       time: 0,
     },
     computed: {
       expensive() {
         console.log("computed run");
         let s = 0;
         for (let i = 0; i < 10000000; i++) s += i;
         return s + this.a + this.b;
       },
     },
     methods: {
       calculate() {
         this.time = Date.now();
       },
     },
   });
 </script>
```

Based on the above data, answer the given subquestions.

9. [Source Q29; MSQ]

What are the benefits of computed caching in Vue?

(a) Reduced CPU usage

(b) Faster rendering

(c) Avoids unnecessary recomputation

(d) Makes all computations asynchronous

[Source ans: a, b, c]

2026 May06 ET

10. [Source Q13; MCQ]

What happens when this Flask caching code runs?

```
from flask import Flask
from flask_caching import Cache
import time

app = Flask(__name__)
app.config['CACHE_TYPE'] = 'SimpleCache'
app.config['CACHE_DEFAULT_TIMEOUT'] = 300

cache = Cache(app)

@cache.memoize(timeout=300)
def compute_data(user_id):
    time.sleep(5)
    return f"Data for user {user_id}"

@app.route('/api/info/<int:user_id>')
def get_info(user_id):
    return compute_data(user_id)
```

If requests for user IDs 1, 2, 1 are made within 5 minutes, what's the total response time?

(a) 15 seconds

(b) 10 seconds

(c) 5 seconds

(d) 0 seconds

[Source ans: b]

11. [Source Q17; MSQ]

Which of the following is the correct statement?

(a) Cookie size can impact the website performance.

(b) Cookie size does not affect performance in any practical/meaningful sense.

(c) A request to static.example.com will include cookies set for example.com.

(d) A request to static.example.com will not include cookies set for example.com.

[Source ans: b, c]

12. [Source Q22; MCQ]

Consider the flask application app.py and an HTML file index.html:

app.py:

```
from flask import Flask, jsonify
from datetime import datetime
import time

app = Flask(__name__)


@app.route('/')
def home():
    time.sleep(10)
    return jsonify(datetime.now().second), 200,
{'Access-Control-Allow-Origin': '*'}


@app.route('/profile')
def profile():
    time.sleep(30)
    return jsonify(datetime.now().second), 200,
{'Access-Control-Allow-Origin': '*'}


if __name__ == "__main__":
    app.run(debug=True)
```

If a user visit index.html at 10:10:10 AM, what will be logged in the console?

index.html:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Document</title>
  </head>
  <body>
    <script>
      function test() {
        const res1 = fetch('http://127.0.0.1:5000')
        const res2 = fetch('http://127.0.0.1:5000/profile')
        res1
          .then((res) => {
            return res.json()
          })
          .then((data) => {
            console.log(data)
          })

        res2
          .then((res) => {
            return res.json()
          })
          .then((data) => {
            console.log(data)
          })
      }
      test()
    </script>
  </body>
</html>
```

(a) 20, 50

(b) 20, 40

(c) 40, 40

(d) 50, 50

[Source ans: b]

2026 Sep13 ET

13. [Source Q9; MCQ]

Match the performance metric in Column I with the most appropriate description in Column II.

Column I	Column II
1. First Contentful Paint (FCP)	A. Measures unexpected movement of page elements
2. Largest Contentful Paint (LCP)	B. Measures how long the page is blocked from responding to user interaction
3. Total Blocking Time (TBT)	C. Measures when something first appears on the screen
4. Cumulative Layout Shift (CLS)	D. Measures when the largest visible content is rendered

(a) 1-C, 2-D, 3-B, 4-A

(b) 1-B, 2-C, 3-A, 4-D

(c) 1-C, 2-B, 3-D, 4-A

(d) 1-A, 2-B, 3-D, 4-C

[Source ans: a]

14. [Source Q17; MSQ]

Which of the following are valid capabilities of Lighthouse while analyzing a website?

(a) Measuring performance metrics such as FCP, LCP and TBT

(b) Simulating slow networks and throttled conditions

(c) Automatically fixing accessibility and SEO issues in the source code

(d) Emulating mobile and desktop devices

[Source ans: a, b, d]